

Content-Provenance Watermarking for AI-Generated Media

*Submitted in fulfillment for the Internal
Assessment of
Multimedia Systems (BITE314L)*

in

B.Tech. - Information Technology

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Fall Semester 2026-27

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1. ABSTRACT

The rapid growth of Generative Artificial Intelligence has transformed the way digital content is created, making it possible to generate highly realistic images, videos, and other multimedia within seconds. While these advancements have opened new opportunities for creativity and automation, they have also introduced serious concerns regarding content authenticity. AI-generated media can be modified, redistributed, or manipulated with ease, making it increasingly difficult to determine its original source. This has contributed to the growing spread of misinformation, deepfakes, copyright violations, and digital impersonation.

This project, **"Content-Provenance Watermarking for AI-Generated Media,"** proposes an integrated multimedia framework that enables reliable verification of AI-generated images by embedding invisible provenance information directly into the media. Instead of relying solely on editable metadata, the system combines robust DWT-DCT watermarking, cryptographic hashing, and provenance registration to establish a secure chain of ownership.

The proposed framework utilizes React and Tailwind CSS for the user interface, FastAPI for backend services, PostgreSQL for secure provenance storage, OpenCV and Pillow for multimedia processing, and Stable Diffusion with GPT APIs for AI content generation. The embedded watermark is designed to remain imperceptible while surviving common image transformations such as compression, resizing, and minor edits.

By integrating multimedia processing, security mechanisms, and AI technologies into a single platform, the proposed system aims to improve trust, transparency, and accountability within AI-generated media ecosystems.

2. INTRODUCTION

Artificial Intelligence has fundamentally changed digital content creation. Modern generative models such as Stable Diffusion can produce highly realistic images from simple text prompts, making AI-generated media widely accessible to individuals, businesses, researchers, and content creators.

Although these technologies offer remarkable creative possibilities, they have also made it increasingly difficult to distinguish authentic media from manipulated or AI-generated content. Images can be edited, redistributed across multiple platforms, or intentionally altered while losing important ownership information stored in conventional metadata. As a result, verifying the authenticity of digital media has become a major challenge.

Content provenance refers to the ability to identify where digital content originated, who created it, and whether it has been modified since its creation. A reliable provenance

framework not only protects creators but also helps organizations verify authenticity before trusting or sharing digital content.

This project addresses these challenges by developing a unified multimedia system that embeds invisible provenance information into AI-generated images. Unlike traditional approaches that depend solely on metadata or visible watermarks, the proposed framework integrates watermarking, cryptographic verification, and secure provenance storage to maintain authenticity throughout the media lifecycle.

2.1 BACKGROUND

The rapid increase in AI-generated content has transformed digital communication. Every day, millions of AI-generated images are shared across websites, social media platforms, news portals, and enterprise applications. While this democratizes creativity, it also creates new challenges in identifying genuine content.

Traditional approaches for proving ownership generally rely on metadata or visible watermarks. Unfortunately, metadata can be removed during image sharing, while visible watermarks can often be cropped or edited without much difficulty. Consequently, these approaches fail to provide reliable authentication under real-world conditions.

Recent developments in multimedia security have introduced invisible watermarking techniques, where ownership information is embedded directly into image pixels without affecting visual quality. Signal-processing methods such as Discrete Wavelet Transform (DWT) and Discrete Cosine Transform (DCT) have shown promising results because they allow hidden information to survive compression and moderate editing operations.

At the same time, cryptographic hashing techniques such as SHA-256 provide an additional layer of integrity verification by generating unique digital fingerprints for media files. Combining these techniques with secure database management enables the creation of an end-to-end provenance framework capable of preserving authenticity throughout the media lifecycle.

2.2 MOTIVATION

The motivation behind this project comes from the growing misuse of AI-generated media. As generative models become more powerful, distinguishing authentic images from manipulated content becomes increasingly difficult. Fake news, deepfakes, unauthorized content redistribution, and copyright disputes are now common challenges across digital platforms.

Existing verification methods often depend on metadata that can be removed or modified during sharing. Similarly, visible watermarks negatively affect image aesthetics while

remaining vulnerable to editing and cropping.

The proposed system aims to overcome these limitations by embedding provenance information directly into the multimedia content itself. This approach enables ownership verification even after common image transformations, thereby improving trust and accountability in AI-generated media.

Beyond security, the project also demonstrates how multimedia processing, artificial intelligence, image watermarking, and cryptographic techniques can be integrated into a practical real-world application that supports creators, organizations, and digital platforms.

3. LITERATURE SURVEY

Researchers have proposed several techniques to improve the authenticity and ownership verification of digital media. Each method contributes to solving part of the problem; however, none provides a complete end-to-end provenance framework.

Traditional visible watermarking methods display ownership information directly on the image. Although effective for discouraging unauthorized usage, these watermarks are often removed through cropping or image editing.

Invisible watermarking techniques embed hidden information within image pixels. These methods preserve visual quality but frequently struggle to maintain robustness after JPEG compression, resizing, or filtering operations.

Metadata-based approaches store ownership information externally using formats such as EXIF or emerging standards like C2PA. While structured and easy to manage, metadata can be stripped during file conversion or sharing across platforms.

Blockchain-based provenance systems provide tamper-resistant ownership records through distributed ledgers. However, these systems generally introduce additional computational costs, increased latency, and deployment complexity.

Consequently, existing solutions solve individual problems but rarely combine watermarking, secure verification, multimedia processing, and provenance management into a unified framework.

3.1 SUMMARY OF TECHNIQUES USED

S.No	Category	Technology / Technique	Purpose in Project
1	Frontend	React, Tailwind CSS	User interface for interacting with the provenance dashboard and generation tools.
2	Backend	FastAPI, Python	Handling high-performance API requests, authentication, and logic processing.
3	AI & Generation	Stable Diffusion, GPT APIs	Generating AI media content that requires provenance tracking.
4	Security & DB	SHA-256, PostgreSQL	Secure hashing for signatures and relational database management for provenance registration.
5	Multimedia Tech	OpenCV, FFmpeg, Pillow, DWT-DCT	Robust neural watermarking and fundamental image/video processing operations.

3.2 RESEARCH GAP

A detailed study of existing techniques reveals several limitations. Most provenance solutions focus on only one component, such as watermarking, metadata management, or blockchain verification. Since these components operate independently, organizations often require multiple disconnected tools to establish complete authenticity verification.

Some invisible watermarking techniques fail to withstand common image modifications such as compression, resizing, or filtering. Likewise, metadata-based approaches remain vulnerable because metadata can easily be removed during content distribution.

The proposed project bridges these limitations by integrating watermark embedding, secure hashing, provenance registration, and automated verification into a single multimedia framework capable of supporting AI-generated content.

4. PROBLEM STATEMENT

AI-generated media is increasingly being shared across digital platforms without reliable mechanisms for verifying authenticity or ownership. Conventional metadata is often removed during image distribution, while existing watermarking techniques frequently fail after common image editing operations.

As a result, creators cannot reliably prove ownership, organizations struggle to verify authenticity, and users find it difficult to distinguish genuine content from manipulated media. Therefore, there is a need for a comprehensive multimedia framework capable of embedding robust provenance information directly into AI-generated images while maintaining image quality and enabling secure verification throughout the content lifecycle.

5. OBJECTIVES

The proposed project is designed with the following objectives:

Primary objective

To develop a secure and transparent multimedia framework for embedding and verifying provenance information within AI-generated media.

Specific Objectives

- Embed invisible provenance watermarks into AI-generated images.
- Preserve visual quality after watermark embedding.
- Ensure watermark robustness against compression, resizing, and editing.
- Generate secure cryptographic signatures using SHA-256.

- Store provenance information within PostgreSQL.
- Verify ownership using automated verification APIs.
- Provide a user-friendly web interface for content registration and verification.

6. METHODOLOGY & DESIGN

The proposed system architecture implements a stepwise pipeline with strict verification at both publication and retrieval stages. The workflow is designed to seamlessly integrate with standard content distribution pipelines:

Step 1 – AI Content Generation

Images are generated using AI models such as Stable Diffusion.



Step 2 – Feature Analysis

The system analyzes image characteristics and identifies suitable embedding regions that maximize robustness while maintaining visual quality.



Step 3 – Watermark Embedding

A DWT-DCT based watermark is invisibly embedded within the generated image.



Step 4 – Secure Hash Generation

A SHA-256 hash is computed to uniquely identify the watermarked image.



Step 5 – Provenance Registration

Ownership details, cryptographic hash, and generation metadata are securely stored in PostgreSQL.



Step 6 – Verification and Authenticity Check

Whenever the media is shared or retrieved, the embedded watermark is extracted and verified against the stored provenance record. Any mismatch indicates possible tampering.

7. EXPECTED OUTCOMES & FUTURE WORK

The successful implementation of the proposed framework is expected to establish a reliable and transparent mechanism for verifying the authenticity of AI-generated media. By embedding invisible provenance watermarks and combining them with cryptographic hashing and secure provenance registration, the system will enable creators and organizations to prove ownership while maintaining the original visual quality of images. The framework is designed to withstand common image transformations such as compression, resizing, and minor editing, ensuring that provenance information remains intact throughout the media lifecycle. In addition, the project aims to reduce the spread of manipulated or misleading AI-generated content by providing an efficient verification mechanism, thereby promoting greater trust, accountability, and copyright protection in digital media ecosystems.

Looking ahead, the proposed system offers significant opportunities for further enhancement and real-world deployment. Future work can focus on extending the framework to support videos, audio files, and other multimedia content, creating a comprehensive provenance solution across multiple media formats. The system can also be integrated with emerging standards such as C2PA to improve interoperability across digital platforms. Incorporating blockchain-based provenance records could provide an additional layer of tamper-resistant ownership verification, while cloud-based verification services would enable scalable deployment for large organizations. Furthermore, browser extensions and mobile applications could allow users to instantly verify the authenticity of online content. With these enhancements, the framework has the potential to be adopted in social media platforms, journalism, digital art marketplaces, legal evidence management, government communications, educational institutions, and enterprise AI systems where trustworthy AI-generated content is becoming increasingly important.

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